

Module 1

A PREVIEW OF CALCULUS

In this section in your text, the author presents an overview of calculus. Read it as a story that exposes you to some of the ideas in calculus that will be covered in more detail later. The major thread presented in the five parts is the concept of a limit. The hope is that by seeing the use of limits in seemingly different situations, you will have a broader base to approach Chapter 2. If you have questions on this section, include them in your first assignment.

SECTION 1.1: FOUR WAYS TO REPRESENT A FUNCTION

NOTATION

Understanding the concept of function is central to the study of calculus. In simple form, a function has an input (domain) and an output (range). In the notation, $f(x)$, x is an input value while $f(x)$ is an output value. If $f(x) = x^2$, then $f(2) = 4$; 2 is in the domain of the function while 4 or $f(2)$ is in the range.

In **Figure 4**, note that $f(2)$ represents the length of a line segment. This is a graphical or geometric interpretation of the notation and will be significant in visualizing algebraic relationships.

DOMAINS

For the types of functions considered in this section of your text, there are two restrictions on the domain:

1. A denominator cannot equal zero.
2. An algebraic expression under a square root sign must be positive or zero.

For $f(x) = \frac{1}{x-3}$, the denominator $x - 3$ equals 0 if $x = 3$. The domain is the set of all real numbers except $x = 3$. This can be abbreviated to, domain: all $x \neq 3$.

For $f(x) = \frac{x}{x^2-1}$, the domain is all $x \neq \pm 1$.

The second restriction appears in $f(x) = \sqrt{x-2}$. Now $x-2$ must be positive or zero. Hence $x-2 \geq 0$, which leads to the domain, all $x \geq 2$.

A different situation arises for $f(x) = \sqrt{x^2-4}$. The domain is determined by the quadratic inequality $x^2-4 \geq 0$ or $(x+2)(x-2) \geq 0$. The methods for determining the solution to this type of inequality are discussed in **Appendix A, Examples 3 and 4**. Either method will lead to the solution,

$$x \leq -2 \text{ or } x \geq 2.$$

Basically, $(x+2)(x-2) \geq 0$ is true if *both* factors are positive (or zero) or *both* factors are negative. Then the product is positive or zero. A common mistake that is made in solving this type of inequality is to think that one can look at only one factor, say $x+2$, and find a solution. If $x = -1$, then $x+2$ is positive ($-1+2 = 1$), but the other factor, $x-2$, is negative ($-1-2 = -3$). Hence the product, $(x+2)(x-2)$, is negative and doesn't satisfy the condition that $(x+2)(x-2)$ be positive or zero. You may have another method for finding the solution but make sure that it works.

RANGE

Determining the range of a function can be a more challenging task. One method is to graph the function and then read the graph (see **Figure 5**). Because one aspect of calculus relates to more precise graphing methods this will be postponed for later consideration.

REPRESENTATIONS OF FUNCTIONS

A major point in this discussion of the four possible ways to describe a function is that sometimes a function is described by a table of values rather than an explicit formula. In particular, note that **Figure 9** contains a set of discrete points that match a table of values. A table doesn't give the values between the points. Later we will say we do not have a *continuous* function. The curve drawn in **Figure 10** is an approximation that allows for estimates between the points. Also note the first use of the words, *mathematical model*.

WORD PROBLEMS

Example 5, page 12, is the first exercise that illustrates the procedure of translating words to a symbolic form or a function. Note that the diagram is a significant intermediate step. Most students find this type of exercise the most challenging problem in calculus. It may help if you realize that you are not the only student who experiences difficulty in creating a function from a word description. This early appearance means you will have many opportunities to develop this important skill. Please put extra effort into looking at the structure of each word problem. As a start, consider the following:

1. For the cost, \$10 per square meter, square meter refers to area. Finding the area of the base and then the area of the four sides is an essential intermediate step.
2. The first expression for C , $20w^2 + 36wh$, has two variables, w and h . Graphing this would require three axes or three dimensions. This is much more complex and is not covered in the text until Chapter 12. So go back to the problem, and use the fact that the volume is 10 cubic meters.
Volume = length $\times$ width $\times$ height = $w(2w)h = 2w^2h = 10$.

Now the important point is how we use this equation. Solve for h and substitute in $20w^2 + 36wh$.

3. The variable h is replaced by $10/2w^2$ or $5/w^2$, and the resulting function, $20w^2 + 180/w$ has only one variable. Besides noting the use of the diagram and area as connecting links, give special attention to the algebraic change from two variables to one variable. This is an important component in this type of problem.

ODD-EVEN FUNCTIONS

Note that there are two events here: an algebraic process and the geometric or graphical interpretation. In the algebraic process, replace x with $-x$. There are three possible results.

Examples

- | | | | |
|----|---|-----------------|--|
| 1. | No change | $f(-x) = f(x)$ | $f(x) = x^2 + x^4$
$f(-x) = (-x)^2 + (-x)^4$
$= x^2 + x^4 = f(x)$ |
| 2. | Exactly opposite | $f(-x) = -f(x)$ | $f(x) = x^3 + x^5$
$f(-x) = (-x)^3 + (-x)^5$
$= -x^3 + (-x^5)$
$= -(x^3 + x^5) = -f(x)$ |
| 3. | Mixed result
Neither of the
first two results | | $f(x) = x + x^2$
$f(-x) = -x + (-x)^2$
$= -x + x^2$
Not equal to $f(x)$
Not equal to $-f(x)$. |

In the graphical interpretation, consider the following:

1. Assume x is a positive number.
2. Find x and $-x$ on the x -axis. The two points are the same distance from $x = 0$.
3. Then remember these are input values.
4. If $f(x)$ is even, the output or y -values are the same.

$$y = f(-x) = f(x) = y$$

Because this is true for *each* x -value, we have one type of symmetry.

5. If $f(x)$ is odd, the output or y -values are opposites. In **Figure 20**, $f(x)$ is a positive number matching the length of a red line segment with two horizontal marks. Then $-f(x)$ is a negative number matching another red line segment with two horizontal line marks below $-x$. This entire process can be described by saying we have symmetry about the origin.

Finally, these two symmetries are possibilities that you may or may not use in graphing functions. Frequently, other tools are used, and these symmetries are in the background but not brought into focus.

INCREASING AND DECREASING FUNCTIONS

Increasing and decreasing refer to changes in y values *as we move from left to right*. This is easier to see in graphical form, **Figure 22**, than in the algebraic form in the box after **Figure 22**. Read the contents of the box carefully. This algebraic form presents information that can later be used with a fundamental concept of calculus, the derivative. At first the derivative will be presented in an algebraic form that will also have a graphical interpretation. But we can't do everything in a graphical form; hence the need for an algebraic description. For now make sure you can read the algebraic symbols AND see how they relate to the graph.

SECTION 1.2: MATHEMATICAL MODELS

Read **page 22** carefully. Note that a *mathematical model* is a mathematical description of some real-world phenomenon. The description is often a function or an equation. Instead of a function being just a collection of algebraic symbols the hope here is that you will

realize it can be description of some physical event. Then the concept of a function can be more meaningful.

The simplest model is a linear model which means that a linear equation can be used as a description. Note that in **Example 2**, the data consists of a set of points, shown in **Figure 4**. Then two methods for finding the equation of a line that can be used for approximations between the points are discussed in some detail.

In the remainder of this section, there is list of different types of functions. Each can serve as a model but the emphasis is on the form of a particular type along with possible graphs. This list of functions can be significant because of the way the study of calculus is structured. Basic calculus procedures that apply to one entry in the list will not be adequate for another. So as you learn calculus, you will better appreciate the difference between the types of functions. Also as you do more applications (translating words into functions), you may be surprised to discover how a particular type of function fits a given problem.

The *key* to understanding the following list is the *algebraic form* in each category. Also note that the list differs a bit from the one in the text.

Types of Functions	Basic Examples
1. Polynomial	$f(x) = 3x^5 + 4x^2 + 2$
2. Rational	$f(x) = \frac{x+1}{x-2}$
3. Algebraic	$f(x) = \sqrt{x}$
4. Trigonometric	$f(x) = \sin x$
5. Exponential	$f(x) = 2^x$
6. Logarithmic	$f(x) = \log_2 x$
7. Transcendental	$f(x) = \sin^{-1} x + 2^x + \log_2 x$

POLYNOMIAL

The form of a polynomial function is

$$P(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_2 x^2 + a_1 x + a_0$$

where $n = 0, 1, 2, 3, \dots$ and $a_n, a_{n-1}, \dots, a_0$ are real numbers.

The important characteristic is that the exponent n is a positive integer or zero. Then $f(x) = \sqrt{x} = x^{1/2}$ and $f(x) = x^{-1} = 1/x$ are *not* polynomial functions.

Also note that $f(x) = 4$ or $y = 4$ (a horizontal line) fits this category as a special case with degree 0. The text calls this a *constant* function. What the text calls a *power* function, x^a is a polynomial only when a is a positive integer or zero. The form of a *power* function will be significant in the study of calculus.

RATIONAL

The first variation from a polynomial function results when the exponent n is a negative integer. Then $f(x) = x^{-1} = 1/x$ is a basic *rational* function because it fits the form $P(x)/Q(x)$ where P and Q are polynomials.

Special case: $P(x) = \frac{P(x)}{1} = \frac{P(x)}{Q(x)}$ where $Q(x) = 1$. Because $Q(x) = 1$ is a polynomial, *all* polynomial functions are rational functions.

ALGEBRAIC

The next extension allows for fractional exponents or the taking of roots. We are creating a category that includes $f(x) = \sqrt{x} = x^{1/2}$ and $f(x) = \frac{1}{\sqrt[3]{x^2}} = x^{-2/3}$.

The last category, algebraic functions, includes both polynomial and rational functions.

TRIGONOMETRIC

First note that **Appendix D** and the front endpaper provide a review of trigonometry. More information will be given later that indicates which basic trigonometric ideas are essential in the study of calculus. For now note that trigonometric functions are *periodic*: they repeat themselves endlessly and are used to describe vibrations of strings, tides, planetary motion, alternating currents, and other periodic processes.

EXPONENTIAL

This is another different form. For $f(x) = 2^x$, the variable is an exponent that produces a behavior called exponential growth.

LOGARITHMIC

The function $f(x) = \log_2 x$ is related to $f(x) = 2^x$ as the inverse, but again the graphs show different behavior.

These last three types, trigonometric, exponential, and logarithmic, do not fit the form of a polynomial. $\sin x$ and $1 + 2x + 3x^2$ are different forms. (Surprisingly, $\sin x$ can be approximated by polynomial functions, but this comes much later under the topic, Infinite Series, Chapter 11.)

Finally, don't be too concerned if this seems vague. It provides a structure that will be meaningful later in the study of calculus.

GRAPHS OF SECOND DEGREE EQUATIONS

We conclude this section with a list of second degree equations which are covered more extensively in **Appendix C**. Although the study of calculus will provide tools that aid in the graphing of functions, it is important to be able to recognize basic types.

	Name	Examples	Comments
1.	Parabola	$y = 4 - x^2$ $x = y^2 - 4$	opens downward opens to the right
2.	Circle	$x^2 + y^2 = 4$ $(x - 3)^2 + (y + 1)^2 = 9$	center (0,0) radius = 2 center (3,-1) radius = 3
3.	Ellipse	$4x^2 + 9y^2 = 36$	graph by finding intercepts (0,2) (3,0)
4.	Hyperbola	$x^2 - y^2 = 1$ $y^2 - x^2 = 4$	intercepts; x $(\pm 1, 0)$ y none intercepts; x none y $(0, \pm 2)$

Suggestion: Given a form like, $y = \sqrt{4 - 4x^2}$, square both sides to get

$$y^2 = 4 - 4x^2,$$

and rewrite as

$$4x^2 + y^2 = 4.$$

Then identify this equation as an ellipse from item 3 in the above list. The graph of the original equation, $y = \sqrt{4 - 4x^2}$, is the top half of an ellipse. (Remember that the square root symbol represents a positive number: $\sqrt{4} = 2$ while $-2 = -\sqrt{4}$. Thus, $\sqrt{4 - 4x^2}$ represents a positive number, and this makes y a positive number.)

In the above list of basic functions also include $f(x) = |x|$ (See **Figure 16, page 29**).

Later we will use some of the above basic equations to illustrate ideas or procedures in calculus. Being able to quickly jot down a graph will be important.

SECTION 1.3: NEW FUNCTIONS FROM OLD FUNCTIONS

The first part of this section in your text, Transformations of Functions, presents general information on graphing functions. If you do not use a rotation, a graph can be shifted in four directions: up, down, to the right, or to the left. To understand the clues that produce these shifts, it is important to distinguish between $f(x) + c$ and $f(x+c)$. In $f(x) + c$, we are adding c to a y -value, which shifts the graph up c units. In $f(x+c)$, we are adding c to an x -value (or changing the input values), which shifts the graph c units to the left. Two similar shifts result from subtracting c .

Finally, note that in $y = -f(x)$ the y -value changes to the opposite value, resulting in a reflection about the x -axis. For $y = f(-x)$, we use the opposite of the x value and get a reflection about the y -axis. These are shown in **Figure 2**.

COMPOSITE FUNCTIONS

This method of combining functions will be significant in calculus. Basically, we “drop” one function “into” another function to get a new result. Or we can break a function like $(x^3 + 4)^5$ into two parts, x^5 and $x^3 + 4$. Then $f[g(x)] = (x^3 + 4)^5$ if $f(x) = x^5$ and $g(x) = x^3 + 4$.

We temporarily skip sections 1.4 and 1.5. You can read about Graphic Calculators and Computers at www.stewartcalculus.com